Group -I crossectioncross-section derivation

Corrected copy of the revised Group I file, with all Group I terms. New or changed text is in blue, removed text in red. So that blue marks only the corrections, the colour marks of your file were changed: your blue (the log $\frac{\vee}{\Lambda}$ terms) is now teal, your green $\frac{1}{\Lambda}$ terms are red (they cancel, see the end of Row III), orange is unchanged, and the [Done] marks are gray. The reasons and the numerical checks are in `Group_I_v2_check.pdf` (R1–R5) and `Group_I_check.pdf` (G1–G6). As in your file, $\Lambda \to 0$ wherever possible. The equation numbers are those of your file. A summary of all terms and of the cancellations inside Group I is at the end.

1 Group I (with c.c.)

1.1 Row I [Done ✓]

Only 4 ρ's

1.2 Row II [Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[U^{ab}(w)\mathbb{A}_i^{(3)\,b}(k^+,-w) - \bar{\mathbb{A}}_i^{(3)\,a}(k^+,-w)\right]$$

We write it in two parts:

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times \left[\frac{11}{3}+\log\left[\frac{\Lambda}{k^+}\right]-\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}-1\right]\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log\left[1-\frac{k^+}{\vee}\right]^2\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\times \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right] \tag{1.1}$$

RewwritingRewriting, we get

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'} e^{-ik(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon}-2\gamma-\log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times \left[\frac{11}{3}+\log\left[\frac{\vee}{k^+}\right]-2\,\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}-1\right]\right]-\frac{67}{9}+\frac{\pi^2}{3}+\log\left[1-\frac{k^+}{\vee}\right]^2\log^2\left[1-\frac{k^+}{\vee}\right]+\log\left[1-\frac{k^+}{\vee}\right]\log\left[\frac{\vee}{k^+}\right]\right]\times \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^b(x)-U^{ab}(x)\,\rho^b(x)\right] \tag{1.2}$$

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{g^4}{64\pi^6}\frac{1}{k^+}\int_{w,w',x'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y} e^{ik\cdot(w-y)}e^{ip\cdot(y-x)}\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\left[\left[-\log\left(\frac{\vee}{\Lambda}\right)+\log\left[\frac{\vee}{k^+}\right]\right]\left[\frac{k^2p^i}{(k-p)^2}-\frac{k\cdot p\,k^i}{(k-p)^2}\right]-2\,k^i\,\log\left(\frac{\vee}{\Lambda}\right)-2k^i\,\log\left(\frac{k^+p^2}{\vee\,k^2+k^+p^2}\right)+\frac{(2p^2-2\,k\cdot p+k^2)\,k^i+k^2p^i}{(k-p)^2}\log\left[1+\left(\frac{\vee}{k^+}-1\right)\frac{k^2}{(k-p)^2}\right]-\frac{(p^2\,k^i+k^2\,p^i)}{(k-p)^2}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]-k^i\,\log\left[\frac{\vee}{k^+}\right]\right]\times \left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{hbc}\,U^{ah}(w)\,\{\rho^b(\mathbf{x}),\,\rho^c(\mathbf{y})\}-f^{abc}\,U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(\mathbf{x}),\,\rho^e(\mathbf{y})\}\right] \tag{1.3}$$

$$\frac{d^3N_{2\,\rm NLO}}{d^3k} = \frac{d^3N_{2\,a}}{d^3k}+\frac{d^3N_{2\,b}}{d^3k} \tag{1.4}$$

$$\frac{d^3N_{2\,a}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{i\,g^4}{32\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{\Lambda}\right) - \log\left(\frac{\vee}{k^+}\right)\right]\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x'}\frac{(x'-w')^i}{(x'-w')^2}\,\mathfrak{F}^i(w,x,y)$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]$$
$$\times\Big[\left[\log\left(\frac{\vee}{\Lambda}\right)\mathfrak{F}^i_\beta(w,x,y) - \log\left(\frac{\vee}{k^+}\right)\mathfrak{F}^i(w,x,y)\right]U^{ah}(w)\,f^{hbc}\left\{\rho^b(x),\rho^c(y)\right\}$$
$$-\left[\log\left(\frac{\vee}{\Lambda}\right) - \log\left(\frac{\vee}{k^+}\right)\right]\mathfrak{F}^i(w,x,y)\,f^{abc}\,U^{bd}(x)\,U^{ce}(y)\left\{\rho^d(x),\rho^e(y)\right\}\Big] \tag{1.5}$$

with

$$\mathfrak{F}^i(w,x,y) = \frac{(x-w)^i}{(x-w)^2}\int_z\frac{(w-z)\cdot(y-z)}{(w-z)^2(y-z)^2} - \frac{(y-w)^i}{(y-w)^2}\int_z\frac{(w-z)\cdot(x-z)}{(w-z)^2(x-z)^2}$$
$$-\left[\frac{(x-w)^i}{(x-w)^2} - \frac{(y-w)^i}{(y-w)^2}\right]\int_z\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2} \tag{1.6}$$
$$= -\pi\left[\frac{(x-w)^i}{(x-w)^2}\log\frac{(w-y)^2}{(x-y)^2} - \frac{(y-w)^i}{(y-w)^2}\log\frac{(w-x)^2}{(x-y)^2}\right]. \tag{1.7}$$

Here $\mathfrak{F}^i_\beta$ is the first line of $\mathfrak{F}^i$, and $\mathfrak{F}^i_\alpha = \mathfrak{F}^i - \mathfrak{F}^i_\beta$ its second line:

$$\mathfrak{F}^i_\beta(w,x,y) = \frac{(x-w)^i}{(x-w)^2}\int_z\frac{(w-z)\cdot(y-z)}{(w-z)^2(y-z)^2} - \frac{(y-w)^i}{(y-w)^2}\int_z\frac{(w-z)\cdot(x-z)}{(w-z)^2(x-z)^2},$$
$$\mathfrak{F}^i_\alpha(w,x,y) = -\left[\frac{(x-w)^i}{(x-w)^2} - \frac{(y-w)^i}{(y-w)^2}\right]\int_z\frac{(x-z)\cdot(y-z)}{(x-z)^2(y-z)^2}.$$

Why (G3, R2). `Calculation_main.tex` drops $-[\mathbb{N}^{(2)},\mathbb{A}^{(1)}_2]$ in the unitarity relation for $\mathbb{A}^{(3)\dagger}_1$. The two operators do not commute: $\left[\int_{x,y}K(x,y)\rho^b(x)\rho^b(y),\rho^a(u)\right] = if^{bac}\int_x K(x,u)\{\rho^b(x),\rho^c(u)\}$.

- The $-2k^i\log\frac{\vee}{\Lambda}$ term of the bracket in (1.3) is exactly this commutator; in coordinate space it is the $\mathfrak{F}_\alpha$ part of $\mathfrak{F}$.
- With the virtual factor ordered in p^+ (gluons softer than k^+ to the left of $\mathbb{A}^{(1)}$, harder ones to the right), the $\log\frac{\vee}{\Lambda}$ part of the commutator drops out of the term with the charges ρ not rotated, $U^{ah}(w)f^{hbc}\{\rho^b(x),\rho^c(y)\}$.
- Only this three-ρ term $\propto\log\frac{\vee}{\Lambda}$ changes. The symmetrized three-ρ terms cancel only with this change (fix F4).
- $\mathfrak{F}_\alpha$ and $\mathfrak{F}_\beta$ are separately logarithmically divergent at large z. The divergent parts are proportional to the total charge.

$$\frac{d^3N_{2\,b}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{64\pi^6}\,\frac{1}{k^+}\int_{w,w'} e^{-ik\cdot(w'-w)}\int d^2p\,d^2k\int_{x,y,x'} e^{ik\cdot(w-y)}\,e^{ip\cdot(y-x)}\,\frac{(x'-w')^i}{(x'-w')^2\,\mathbf{p}^2\mathbf{k}^2}\,R^i(\mathbf{k},\mathbf{p})$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{hbc}\,U^{ah}(w)\left\{\rho^b(x),\rho^c(y)\right\} - f^{abc}\,U^{bd}(x)\,U^{ce}(y)\left\{\rho^d(x),\rho^e(y)\right\}\right] \tag{1.8}$$

where

$$R^i = -2k^i\log\left(\frac{p^2}{k^2+\frac{k^+}{\vee}p^2}\right) + \frac{(2p^2-2\,k\cdot p+k^2)\,k^i + k^2p^i}{(k-p)^2}\log\left[\frac{k^2}{(k-p)^2} + \frac{k^+}{\vee}\left(1-\frac{k^2}{(k-p)^2}\right)\right] - \frac{p^2k^i+k^2p^i}{(k-p)^2}\log\left(1-\frac{k^+}{\vee}\right). \tag{1.9}$$

1.3 Row III

[Done ✓]

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_z\Big[-\bar{\mathbb{A}}^{\dagger\,(1)\,a}_i(k^+,-w') + \mathbb{A}^{\dagger\,(1)\,b'}_i(k^+,-w')U^{b'a}(w')\Big]\times$$
$$\Big[-\bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+,w;p^+,-z)\bar{\mathbb{A}}^{(1)\,b}_j(p^+,-z) + \bar{\mathbb{B}}^{\dagger\,ba}_{3ji}(k^+,w;p^+,-z)U^{bc}(z)\mathbb{A}^{(1)\,c}_j(p^+,-z)\Big]$$

$$
\begin{aligned}
\frac{d^3 N_{\rm NLO}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{w,w'} e^{-ik(w'-w)} \Bigg\{ & \\
& \frac{ig^4\, f^{abd}}{8\pi^4} \int_{k^+ + \Lambda}^{\vee} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2} \frac{(p^+ + k^+)}{(k^+ - p^+)^2} \frac{1}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& + \frac{ig^4\, f^{abd}}{8\pi^4} \int_{\Lambda}^{k^+ - \Lambda} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2} \frac{(p^+ + k^+)}{(k^+ - p^+)^2} \frac{1}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{8\pi^4} \int_{\Lambda}^{\vee} \frac{dp^+}{2\pi} \frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2} \frac{\left[k^+\,(x-w)^2 + p^+\,(x-z)^2\right]}{[p^+\,(x-z)^2 - k^+\,(x-w)^2]} \frac{1}{p^+\,k^+} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{4\pi^4} \frac{1}{k^+} \int_{\Lambda + k^+}^{\vee} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')^i\,(y-z)^j\,(z-w)^k}{(x'-w')^2\,(y-z)^2\,(z-w)^2} \frac{[k^+\,(x-w)^m - p^+\,(x-z)^m]}{[k^+\,(x-w)^2 - p^+\,(x-z)^2]} \frac{1}{(k^+ - p^+)} \\
& \left[\delta_{ik}\,\delta_{jm} - \frac{k^+}{p^+}\,\delta_{jk}\,\delta_{im} - \frac{k^+}{(k^+ - p^+)}\,\delta_{ij}\,\delta_{km}\right] \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \\
& \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& - \frac{ig^4\, f^{abd}}{4\pi^4} \int_{\Lambda}^{k^+ - \Lambda} \frac{dp^+}{2\pi} \int_{x,y,z,x'} \frac{(x'-w')^i\,(y-z)^j\,(z-w)^k}{(x'-w')^2\,(y-z)^2\,\left[k^+\,(x-w) - p^+\,(x-z)\right]^2} \frac{[k^+\,(x-w)^m - p^+\,(x-z)^m]}{[k^+\,(x-w)^2 - p^+\,(x-z)^2]} \frac{p^+}{(k^+ - p^+)} \\
& \left[\delta_{ik}\,\delta_{jm} - \frac{k^+}{p^+}\,\delta_{jk}\,\delta_{im} - \frac{k^+}{(k^+ - p^+)}\,\delta_{ij}\,\delta_{km}\right] \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \\
& \Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \Bigg\}
\end{aligned}
\tag{1.10}
$$

Integrating wrt p^+ and we write it in three parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)} \Bigg\{ \\
&\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'} \frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}+\frac{1}{k^+-\vee}\right] \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left|\frac{\left[\vee\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[(\Lambda+k^+)\,(x-z)^2-k^+\,(x-w)^2\right]}\right| \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{1}{[(x-z)^2-(x-w)^2]}\times\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right] \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{k^+-\Lambda}\right]-\log\left[\frac{\vee}{\Lambda}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-z)^2+(x-w)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\left[(x-w)^2+(x-z)^2\right]}{\left[(x-z)^2-(x-w)^2\right]^2}\times\log\left|\frac{\left[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[\Lambda\,(x-z)^2-k^+\,(x-w)^2\right]}\right| \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&+\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\log\left[\frac{\vee}{\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2} \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\log\left|\frac{\left[k^+\,(x-w)^2-\vee\,(x-z)^2\right]}{\left[k^+\,(x-w)^2-\Lambda\,(x-z)^2\right]}\right| \\
&\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \qquad (1.11)
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ \\
&\frac{ig^4\,f^{abd}}{8\pi^5}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,[(x-w)^2-(x-z)^2]}\left[\frac{1}{(\vee-k^+)}-\frac{1}{\Lambda}\right] \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{\vee}{\Lambda+k^+}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right|\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right|\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right|\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right|\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&+\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right|\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
&-\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right|\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
&\left[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{1.12}
$$

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& \frac{ig^4 f^{abd}}{8\pi^5}\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right|\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right|\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right|\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right|\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right|\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\log\left|\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right|\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\mathcal{A}\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& +\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\mathcal{A}\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right] \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\right]\Bigg\}
\end{aligned}
\tag{1.13}
$$

$$
\begin{aligned}
\frac{d^3N_{3b\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{ & \\
& -\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\,\mathcal{A}\int_{x,y,z,x'}\frac{(x-w)\cdot(x-z)\,(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)}{(x'-w')^2(y-z)^2(z-w)^2(x-z)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& +\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(y-z)\cdot(z-w)\,(x'-w')\cdot(x-w)}{(x'-w')^2(y-z)^2(z-w)^2(x-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big] \\
& -\frac{ig^4 f^{abd}}{16\pi^5}\frac{1}{k^+}\log\left[\frac{\left[k^+(x-w)-(k^+-\Lambda)(x-z)\right]^2}{\left[k^+(x-w)-\Lambda(x-z)\right]^2}\right]\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2\,(z-w)^2} \\
& \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\Bigg\}
\end{aligned}
\tag{1.14}
$$

where

$$
\begin{aligned}
\mathcal{A} = \frac{1}{\sqrt{(x-w)^2(x-z)^2-[(x-w)\cdot(x-z)]^2}}\Bigg\{ & \arctan\left[\frac{(k^+-\Lambda)(x-z)^2-k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2-[(x-w)\cdot(x-z)]^2}}\right] \\
& -\arctan\left[\frac{\Lambda(x-z)^2-k^+(x-w)\cdot(x-z)}{k^+\sqrt{(x-w)^2(x-z)^2-[(x-w)\cdot(x-z)]^2}}\right]\Bigg\}
\end{aligned}
$$

Row III: corrections.

- **The $\frac{1}{\Lambda}$ terms cancel.** The four terms in red in d^3N_1, d^3N_2, d^3N_{3a} (your green terms) multiply the same structure $C \equiv \frac{ig^4 f^{abd}}{8\pi^5}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}\times$ colour. With $D \equiv (x-z)^2-(x-w)^2$ and the overall signs of d^3N_2 $(-)$ and of the denominators,

$$
\frac{C}{D}\left\{\left[\frac{1}{\Lambda}-\frac{1}{\vee-k^+}\right]+\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]+\left[\frac{1}{\vee-k^+}-\frac{1}{\Lambda}\right]-\left[\frac{1}{\Lambda}-\frac{1}{k^+-\Lambda}\right]\right\}=0.
$$

  They are removed. Numerically the sum is zero to 6×10^{-16} of one term.

- **The two red terms of d^3N_2** that you had marked are equal and opposite; they cancel.

- **Principal values (G4, R3).** The integrand of (1.10) has a pole at $p_*^+ = k^+(x-w)^2/(x-z)^2$. When it lies inside the p^+ range the integral is a principal value, and the argument of the logarithm changes sign. Hence $\log[\cdots]\to\log|\cdots|$ in the 3 logarithms of d^3N_1, the 6 of d^3N_2 and the 6 of d^3N_{3a}. $\mathcal{A}$ and the logarithm of d^3N_{3b} need no change.

1.4 Row IV

[Done ✓]

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_z & \Big[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w')+\mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\Big]\times \\
& \Big[U^{ac}(w)U^{bd}(z)\bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\mathbb{B}_{2ji}^{dc}(k^+,-w;p^+,-z)-\bar{\mathbb{A}}_j^{\dagger\,(1)\,b}(p^+,-z)\bar{\mathbb{B}}_{2ji}^{ba}(k^+,-w;\,p^+,-z)\Big]
\end{aligned}
$$

$$
\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\frac{ig^4}{4\pi^4}\int_\Lambda^{\vee-k^+}\frac{dp^+}{2\pi}\int_{x,y,z,w,x',w'} & e^{-ik(w'-w)}\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\frac{1}{[p^+(x-z)^2+k^+(x-w)^2]} \\
& \left[\frac{\delta^{ij}}{2\,(z-w)^2\,(p^++k^+)}\left[(x-z)^2-(x-w)^2\right]+\frac{1}{k^+}\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2}-\frac{(x-z)^j\,(x-w)^i}{2\,(x-w)^2}\right]\right. \\
& \left.+\frac{1}{p^+}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2}\right]\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big] \\
& \Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x)-f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\Big]
\end{aligned}
\tag{1.15}
$$

Integrating w.r.t. p^+

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{ \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2} - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] - \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\log\left[\frac{\vee}{\Lambda+k^+}\right] + \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[1-\frac{k^+}{\vee}\right]\right]\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\left[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^e(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^e(y)\rho^d(x)\right] \tag{1.16}$$

1.5 Row V

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{2}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\int\frac{dp^+}{2\pi}\int_{x,z}\left[-\bar{\mathbb{A}}_i^{\dagger\,(1)\,a}(k^+,-w') + \mathbb{A}_i^{\dagger\,(1)\,b'}(k^+,-w')U^{b'a}(w')\right]\times \left[-\bar{\mathbb{C}}^{\dagger\,bca}_{1jki}(k^+,w-z;p^+,z-x)\bar{\mathbb{B}}^{bc}_{2jk}(k^+-p^+,-z;p^+,-x) + \bar{\mathbb{C}}^{\dagger\,dca}_{1jki}(k^+,w-z;p^+,z-x)\,U^{db}(x)\,U^{ce}(z)\,\mathbb{B}^{be}_{2jk}(k^+-p^+,-z;p^+,-x)\right]$$

(R5) In Row V the phases $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)p^+/k^+}$ (and the q^+ ones) depend on x, z and p^+; they were moved inside $\int dp^+$ and $\int_{x,y,z,\ldots}$, and the unmatched brace $\Big\}$ was removed.

We write it in two parts:

1.5.1 First part: [Done ✓]

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\Bigg\{\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',y'}\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\,\delta^{(2)}\left[w-z+\frac{p^+}{k^+}\,(z-x)\right] \left[\delta_{im}\,\delta_{jk}-\frac{k^+}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{k^+}{(k^+-p^+)}\,\delta_{ij}\,\delta_{km}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\Bigg\} \tag{1.17}$$

Integrating w.r.t. w, we get

$$\frac{d^3N_{1\,\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\Bigg\{\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\,e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}} \frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \left[\delta_{im}\,\delta_{jk}-\frac{k^+}{p^+}\,\delta_{jm}\,\delta_{ik}-\frac{k^+}{(k^+-p^+)}\,\delta_{ij}\,\delta_{km}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]\Bigg\} \tag{1.18}$$

Using + prescription, we get

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\, \frac{ig^4}{16\pi^4}\,\frac{1}{k^{+\,2}}\int_0^{k^+}\frac{dp^+}{2\pi}\, e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\frac{p^+}{k^+}}$$
$$\frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\left[\delta_{im}\,\delta_{jk} - \left[\frac{k^+}{p^+}\right]_+\delta_{jm}\,\delta_{ik}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times$$
$$\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\, e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\,\frac{ig^4}{16\pi^4}\,\frac{1}{k^+}\int_0^{k^+}\frac{dq^+}{2\pi}\, e^{-ik\cdot(w'-x)}\, e^{-ik\cdot(x-z)\frac{q^+}{k^+}}$$
$$\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}\left[\frac{1}{q^+}\right]_+$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]$$
$$-\frac{1}{(2\pi)^3}\int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2}$$
$$\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right] \tag{1.19}$$

1.5.2 Second part:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik\cdot(w'-w)}\left\{\right.$$
$$\frac{g^4}{8\pi^4}\,\frac{1}{k^{+\,3}}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'}\frac{p^+\,(k^+-p^+)}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\,\delta^{(2)}\left[w-z+\frac{p^+}{k^+}\,(z-x)\right]\times$$
$$\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]-\frac{k^{+\,2}}{(k^+-p^+)^2}\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.$$
$$+\frac{k^+\,\delta_{im}}{p^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]-\frac{k^{+\,2}}{p^{+\,2}}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{k^+\,\delta_{im}}{(k^+-p^+)}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$\left.\left.-\frac{k^{+\,2}}{p^+\,(k^+-p^+)}\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\right.$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[f^{adc}\,f^{ebf}\,U^{db}(x)\,U^{ce}(z)-f^{abc}\,f^{cbd}\,U^{df}(y)\right]\rho^{b'}(x')\,\rho^f(y)\left.\right\} \tag{1.20}$$

Integrating w.r.t. w, we get

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{ \frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\, \frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[f^{adc}\,f^{ebf}\,U^{db}(x)\,U^{ce}(z)-f^{abc}\,f^{cbd}\,U^{df}(y)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.21}$$

This can be written as:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{ \frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\, \frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c U^{af}(y)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \tag{1.22}$$

This term contains a UV divergence at $x \to z$. So, we use the following identity to isolate the UV term and the finite term. The identity as follows:

$$f^{abc}\,f^{def}\,U^{be}(z)\,U^{cf}(z) = N_c\,U^{ad}(z)$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{$$

$$\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$

$$\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]$$

$$+\,\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$-\,\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$

$$+\,\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg]\Bigg\}$$

$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(y)+N_c\,U^{af}(x)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\} \tag{1.23}$$

This can be written in two parts:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{$$

$$\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$

$$\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}} - \frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]$$

$$+\,\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$-\,\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$

$$+\,\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$

$$-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg]\Bigg\}$$

$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\} \tag{1.24}$$

$$
\frac{d^3 N_{2b\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{w'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}} \Bigg\{
$$

$$
\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x'} e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}
$$

$$
\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}-\frac{p^+}{(k^+-p^+)}\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right.
$$

$$
+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]
$$

$$
-\frac{(k^+-p^+)}{p^+}\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]
$$

$$
+\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]
$$

$$
\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}
$$

$$
\left[U^{ab'}(x')-U^{ab'}(w')\right]\,\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\,\rho^{b'}(x')\,\rho^{f}(y)\Bigg\} \tag{1.25}
$$

We will handle the p^+, using + prescription for the first part. We get the following:

$$\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_0^{k^+}\frac{dp^+}{2\pi}\,e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\Bigg\{$$
$$\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}$$
$$+\,\frac{(k^+-p^+)\,\delta_{im}}{k^+}\,\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{k^+}{p^+}\right]_+\,\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]$$
$$+\,\frac{p^+\,\delta_{im}}{k^+}\,\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]$$
$$-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\Bigg]\Bigg\}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\}$$
$$+\,\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}$$
$$\frac{1}{\left[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\left\{\left[\frac{k^+}{q^+}\right]_+\,\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-z)}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'} e^{-ik\cdot(w'-z)}$$
$$\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.$$
$$\left.+\,\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)$$
$$-\,\frac{1}{(2\pi)^3}\int_{w'} e^{-ik\cdot(w'-x)}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'} e^{-ik\cdot(w'-x)}$$
$$\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}$$
$$\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y) \tag{1.26}$$

For the second part, we will first handle the UV divergence and then do plus prescription to handle the p^+.

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k} = \frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} + \frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k}$$

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{\vee}{\Lambda}\right)-2\log\left(\frac{\vee}{k^+}\right)+2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}+\frac{4\Lambda}{k^+}-\frac{\Lambda^2}{k^{+2}}+\frac{2\Lambda^3}{3k^{+3}}\right]$$
$$\times\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}$$
$$\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y). \tag{1.27}$$

Setting $\Lambda \to 0$, we get

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{\vee}{\Lambda}\right) - 2\log\left(\frac{\vee}{k^+}\right) + 2\log\left(1-\frac{\Lambda}{k^+}\right) - \frac{11}{6}\right]\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}$$
$$\left[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[U^{af}(y) - U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y). \quad (1.28)$$

Note (R4). At the accuracy $\Lambda \to 0$ of this equation, $2\log(1-\frac{\Lambda}{k^+})$ is $O(\Lambda/k^+)$, the same order as the dropped $\frac{4\Lambda}{k^+} - \frac{\Lambda^2}{k^{+2}} + \frac{2\Lambda^3}{3k^{+3}}$; likewise $\log\frac{k^+-\Lambda}{\vee} = -\log\frac{\vee}{k^+} + O(\Lambda/k^+)$ everywhere in Row V. Keeping these exact endpoint logarithms is harmless; dropping them gives the bracket $2\log\frac{\vee}{\Lambda} - 2\log\frac{\vee}{k^+} - \frac{11}{6}$.

$$\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{$$
$$e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2 + (k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}$$
$$\times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}} - \frac{p^+}{k^+-p^+}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2} - \frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]$$
$$+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2} + \frac{(y-x)\cdot(y-z)}{2(y-x)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$
$$-\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2} + \frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]$$
$$+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2} - \frac{(y-x)\cdot(y-z)}{2(y-z)^2} - \frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]$$
$$-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2} - \frac{(y-x)^m(y-z)^i}{2(y-z)^2} + \frac{(y-z)^m(x-z)^i}{(x-z)^2} + \frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg]$$
$$\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y) - N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^{f}(y)$$
$$-\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}} + \frac{p^+}{k^+-p^+} + \frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}$$
$$\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y) - N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^{f}(y)\Bigg\}. \quad (1.29)$$

Simplifying, we get

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{\Lambda}^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\Bigg\{ \\
& e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2} \\
& \times\Bigg[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,[(y-x)^2-(y-z)^2]\,\frac{p^+(k^+-p^+)}{k^{+2}} \\
& +\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
& -\frac{k^+-p^+}{p^+}\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right] \\
& +\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right] \\
& -\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\Bigg] \\
& \times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y) \\
& -\frac{1}{k^+}\left[\frac{p^+(k^+-p^+)}{k^{+2}}+\frac{p^+}{k^+-p^+}+\frac{k^+-p^+}{p^+}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2} \\
& \times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\Bigg\} \\
& +\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]} \\
& \frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\frac{k^+}{q^+}\times \\
& \Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(zx)\Big]\rho^{b'}(x')\,\rho^f(y) \\
& -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\;e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]} \\
& \frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right] \\
& \Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(zx)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.30}
$$

(R1) In the two q^+ blocks above (the B_2 term, $-\frac{p^+}{k^+-p^+}=-\frac{k^+}{q^+}+1$) the colour factor is $N_c[U(y)-U(x)]$: the B_2 term belongs to 2b. Only the counterterm has $U(z)$. With this change the equation equals the previous one exactly.

Now, we do plus prescription for $\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k}$ and integrating wrt p^+ and setting Λ to be 0 wherever possible. We get the

following

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{(k^+-\Lambda)}{\vee}\right)\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{(k^+-\Lambda)}{\vee}\right)\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{(k^+-\Lambda)}{\vee}\right)\right]\right]\\
&\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.31}
$$

Now, we write the final answer for Row V:

First part

$$
\begin{aligned}
\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3 k} = \frac{1}{(2\pi)^3} \int_{x,z,y,x',y',w'} & e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\, \frac{ig^4}{16\pi^4}\, \frac{1}{k^{+\,2}} \int_0^{k^+} \frac{dp^+}{2\pi}\, e^{-ik\cdot(w'-z)}\, e^{-ik\cdot(z-x)\frac{p^+}{k^+}} \\
& \frac{(x'-w')^i\,(z-x)^m\,(y-z)^k\,(y'-x)^j}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
& \left[\delta_{im}\,\delta_{jk} - \left[\frac{k^+}{p^+}\right]_+ \delta_{jm}\,\delta_{ik}\right] \times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right] \times \\
& \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right] \\
& - \frac{1}{(2\pi)^3} \int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\, e^{-ik\cdot(x-z)\frac{q^+}{k^+}}\, \frac{ig^4}{16\pi^4}\, \frac{1}{k^+} \int_0^{k^+} \frac{dq^+}{2\pi}\, e^{-ik\cdot(w'-x)}\, e^{-ik\cdot(x-z)\frac{q^+}{k^+}} \\
& \frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \left[\frac{1}{q^+}\right]_+ \\
& \times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right] \times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right] \\
& - \frac{1}{(2\pi)^3} \int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-z)}\, \frac{ig^4}{32\pi^5}\, \frac{1}{k^+} \left[\log\left[\frac{\vee}{\Lambda}\right] + \log\left[\frac{k^+-\Lambda}{\vee}\right]\right] \frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
& \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right] \times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right] \\
& - \frac{1}{(2\pi)^3} \int_{x,z,y,x',y',w'} e^{-ik\cdot(w'-x)}\, \frac{ig^4}{32\pi^5}\, \frac{1}{k^+} \left[\log\left[\frac{\vee}{\Lambda}\right] + \log\left[\frac{k^+-\Lambda}{\vee}\right]\right] \frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\
& \times \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right] \times \left[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^e(y),\rho^b(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^d(y),\rho^e(y')\}\right]
\end{aligned}
\tag{1.32}
$$

Second part

$$
\begin{aligned}
\frac{d^3N_{2a\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\int_0^{k^+}\frac{dp^+}{2\pi}\,e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\Bigg\{\\
&\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\,\frac{p^+}{k^+}}\,\frac{1}{\left[p^+\,(y-x)^2+(k^+-p^+)\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\,\left[(y-x)^2-(y-z)^2\right]\,\frac{p^+\,(k^+-p^+)}{k^{+\,2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2\,(y-x)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]+\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2\,(y-z)^2}-\frac{\left[(y-x)^2-(y-z)^2\right]}{2\,(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m\,(x-z)^i}{(x-z)^2}-\frac{(y-x)^m\,(y-z)^i}{2\,(y-z)^2}+\frac{(y-z)^m\,(x-z)^i}{(x-z)^2}+\frac{(y-x)^i\,(y-z)^m}{2\,(y-x)^2}\right]\right]\Bigg\}\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\Bigg\}\\
&+\frac{1}{(2\pi)^3}\int_{k^+}^0\frac{dq^+}{2\pi}\,e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,e^{-ik\cdot(x-z)\,\frac{q^+}{k^+}}\\
&\frac{1}{\left[(k^+-q^+)\,(y-x)^2+q^+\,(y-z)^2\right]}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\\
&\left\{\left[\frac{k^+}{q^+}\right]_+\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]-\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\right\}\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\int_{w'}e^{-ik\cdot(w'-z)}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}e^{-ik\cdot(w'-z)}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right.\\
&\left.+\frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\int_{w'}e^{-ik\cdot(w'-x)}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left[\frac{\vee}{\Lambda}\right]+\log\left[\frac{k^+-\Lambda}{\vee}\right]\right]\int_{x,y,z,x'}e^{-ik\cdot(w'-x)}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2}\\
&\left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2}-\frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z)+N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.33}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\left[2\log\left(\frac{\vee}{\Lambda}\right)-2\log\left(\frac{\vee}{k^+}\right)+2\log\left(1-\frac{\Lambda}{k^+}\right)-\frac{11}{6}\right]\int_{y,z,x',w'}e^{-ik\cdot(w'-z)}\\
&\left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[U^{af}(y)-U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y).
\end{aligned}
\tag{1.34}
$$

$$
\begin{aligned}
\frac{d^3N_{2b\,\text{finite NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_\Lambda^{k^+-\Lambda}\frac{dp^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{p^+}{k^+}}\,\frac{1}{[p^+(y-x)^2+(k^+-p^+)(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\times\left[\frac{\delta^{im}\,d_\perp}{2\,(x-z)^2}\left[(y-x)^2-(y-z)^2\right]\frac{p^+(k^+-p^+)}{k^{+2}}\right.\\
&+\frac{(k^+-p^+)\,\delta_{im}}{k^+}\left[\frac{(y-z)\cdot(x-z)}{(x-z)^2}+\frac{(y-x)\cdot(y-z)}{2(y-x)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&-\left[\frac{k^+}{p^+}\right]_+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]+\left[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\right]\\
&+\frac{p^+\,\delta_{im}}{k^+}\left[\frac{(y-x)\cdot(x-z)}{(x-z)^2}-\frac{(y-x)\cdot(y-z)}{2(y-z)^2}-\frac{(y-x)^2-(y-z)^2}{2(x-z)^2}\right]\\
&\left.-\left[\frac{(y-x)^m(x-z)^i}{(x-z)^2}-\frac{(y-x)^m(y-z)^i}{2(y-z)^2}+\frac{(y-z)^m(x-z)^i}{(x-z)^2}+\frac{(y-x)^i(y-z)^m}{2(y-x)^2}\right]\right]\\
&\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+}^0\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\left[\frac{k^+}{q^+}\right]_+\times\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^4}\,\frac{1}{k^+}\int_{k^+-\Lambda}^{\Lambda}\frac{dq^+}{2\pi}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,e^{-ik\cdot(z-x)\frac{(k^+-q^+)}{k^+}}\,\frac{1}{[(k^+-q^+)(y-x)^2+q^+\,(y-z)^2]}\\
&\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2}\left[\frac{(y-x)^i(x-z)^m}{(x-z)^2}-\frac{(y-x)^i(y-z)^m}{2(y-z)^2}\right]\\
&\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{(k^+-\Lambda)}{\vee}\right)\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{(k^+-\Lambda)}{\vee}\right)\right]\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(x)\right]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[-\frac{11}{6}+2\left[\log\left(\frac{\vee}{\Lambda}\right)+\log\left(\frac{(k^+-\Lambda)}{\vee}\right)\right]\right]\\
&\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\left[U^{ab'}(x')-U^{ab'}(w')\right]\left[N_c\,U^{af}(y)-N_c\,U^{af}(z)\right]\rho^{b'}(x')\,\rho^f(y)
\end{aligned}
\tag{1.35}
$$

Summary: all Group I terms

Equation numbers as above. $E(r)\equiv\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu_{\overline{MS}}^2}{4}$.

row	term	final form
Row I	four ρ's	not treated here
Row II	part 1 (two ρ)	(1.2)
	part 2a (three ρ)	(1.5), with the G3 change
	part 2b (three ρ)	(1.8) with R^i of (1.9), $\int_{x'}$ added
Row III	d^3N_1, d^3N_2, d^3N_{3a}, d^3N_{3b}	(1.11)–(1.14) with $\mathcal{A}$; $\frac{1}{\Lambda}$ terms removed, $\log\lvert\cdot\rvert$
Row IV		(1.16)
Row V	part 1	(1.19) = (1.32)
	part 2a	(1.26) = (1.33)
	part 2b, UV	(1.28) = (1.34)
	part 2b, finite	(1.31) = (1.35)

Cancellations inside Group I.

1. $\frac{1}{\Lambda}$ **terms (Row III):** they cancel exactly (end of Row III).

2. **Running coupling (Row II part 1 against Row V 2b UV):** the β_0 terms are

$$\left.\frac{d^3N_{\rm II}}{d^3k}\right|_{\beta_0} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{32\pi^4}\frac{1}{k^+}\frac{11}{3}\int_{w,w',x,x'} e^{-ik\cdot(w'-w)}\frac{(x'-w')\cdot(x-w)}{(x'-w')^2(x-w)^2}\big[-E(x-w)\big]$$
$$\times\big[U^{ab'}(x')-U^{ab'}(w')\big]\rho^{b'}(x')\big[U^{ab}(w)-U^{ab}(x)\big]\rho^{b}(x),$$
$$\left.\frac{d^3N_{\rm V}}{d^3k}\right|_{\beta_0} = \frac{1}{(2\pi)^3}\frac{g^4N_c}{16\pi^4}\frac{1}{k^+}\Big(-\frac{11}{6}\Big)\int_{y,z,x',w'} e^{-ik\cdot(w'-z)}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2}E(y-z)$$
$$\times\big[U^{ab'}(x')-U^{ab'}(w')\big]\big[U^{af}(y)-U^{af}(z)\big]\rho^{b'}(x')\rho^{f}(y).$$

Rename $y\to x$, $z\to w$, $f\to b$ in the second. Then

- the phase, the kernel and E become those of the first;
- $[U(x)-U(w)] = -[U(w)-U(x)]$;
- $\frac{g^4N_c}{16\pi^4}\big(-\frac{11}{6}\big) = -\frac{g^4N_c}{32\pi^4}\frac{11}{3}$.

Hence

$$\left.\frac{d^3N_{\rm II}}{d^3k}\right|_{\beta_0} + \left.\frac{d^3N_{\rm V}}{d^3k}\right|_{\beta_0} = 0$$

per ordering and for the c.c. The pole, γ_E, μ and $\log(x-w)^2$ all cancel. With the exact brackets it holds exactly in Λ.

3. **What is left:**

- The $\log\frac{\nu}{\Lambda}$ terms (teal and orange) are the evolution (JIMWLK/BFKL) terms.
- The remaining poles multiply only rapidity logarithms; they combine with Group II, Row IV.
- The charge renormalization of LO gives the $\frac{\beta_0}{2}\delta(1-\xi)$ of the fragmentation counterterm.
- Row II also contains the constant $-\frac{67}{9}+\frac{\pi^2}{3}$: per ordering this is $\frac{\alpha_s}{2\pi}K\times$LO with $K = N_c\big(\frac{67}{18}-\frac{\pi^2}{6}\big)$, the CMW constant.